STAT 4080/5080, SPRING 2026

NAME_____

SIGNATURE_____

MIDTERM EXAM

Due by 2:30 p.m., Thursday, March 19, 2026

INSTRUCTIONS

1. You are not allowed to communicate in any way with anyone in or outside of this class. Anyone found discussing the exam with other people is considered cheating, and thus all parties involved in the discussion will receive a "0" score on the exam. And all parties found violating the codes of academic honesty will be reported to respective colleges.

2. You should type the answers directly onto a word document. The answers have to be as specific as possible. All the relevant outputs should be copied and pasted onto the same document, with clear indications as to where to find those outputs. You have to show all the work in order to earn full credit. However, please only include relevant outputs.

3. You can choose to use any software packages.

4. The midterm exam is due by **2:30 p.m. on Thursday, March 19, 2026.** Please submit a hard copy of your completed before the deadline. Please type your name and sign your signature on the top of this page when submitting your completed midterm exam.

Problem 1

Four different circuit designs are compared on the amount of noise present ($a = 4$ groups, $n = 5$ per group, balanced CRD).

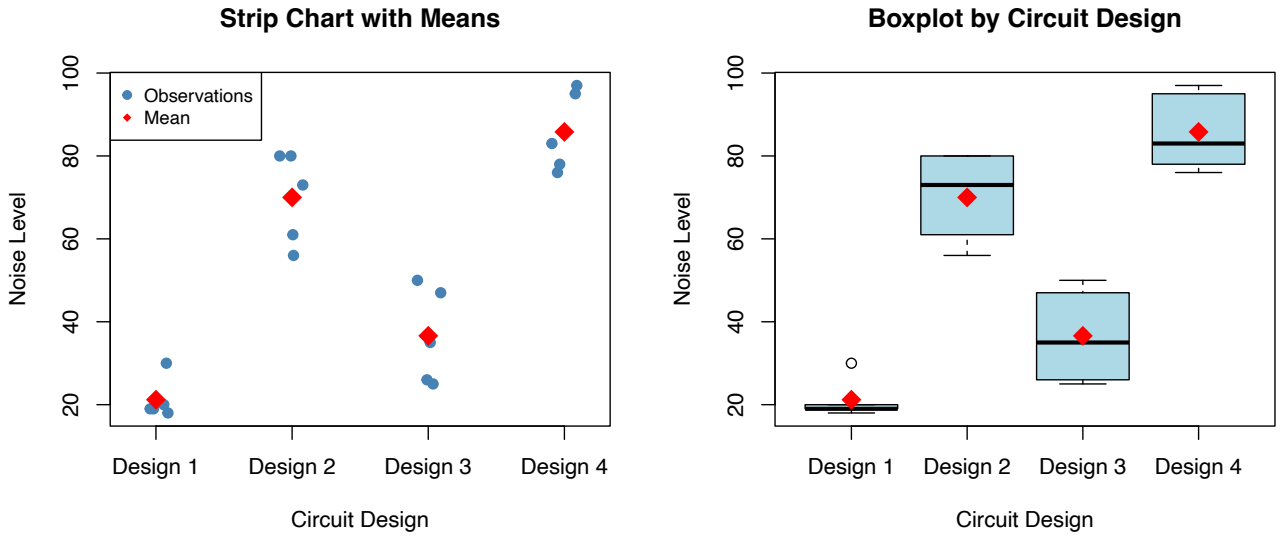
Model (effects model):

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, 3, 4, \; j = 1, \ldots, 5, \quad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$$

Table 1: Summary Statistics — Circuit Noise Data

Circuit	n	Mean	SD	Min	Max
Design 1	5	21.2	4.970	18	30
Design 2	5	70.0	11.023	56	80
Design 3	5	36.6	11.589	25	50
Design 4	5	85.8	9.680	76	97

Part (a) Graphical Comparison of Treatment Means



The strip chart and boxplot reveal clear separation among the four circuit designs. Design 4 consistently produces the highest noise levels (mean 85.8), followed by Design 2 (mean 70.0), then Design 3 (mean 36.6), and Design 1 produces the least noise (mean 21.2). The spreads (variances) appear somewhat larger for Designs 2 and 4, but not drastically heterogeneous. Whereas, Design 1 has a very tight cluster compared to Design 3 or 4.

There appear to be clear differences between the designs. Design 1 and 4 are noticeably at opposite ends of the spectrum, suggesting a significant treatment effect.

Part (b) One-Way ANOVA F-Test

Hypotheses:

$$H_0 : \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0 \quad \text{(equivalently, } \mu_1 = \mu_2 = \mu_3 = \mu_4\text{)}$$

$$H_a : \text{At least one } \tau_i \neq 0$$

Table 2: One-Way ANOVA Table — Circuit Noise

Source	DF	SS	MS	F_value	P_value
Treatment (Circuit)	3	13222.0	4407.333	47.1121	0
Error	16	1496.8	93.550	NA	NA
Total	19	14718.8	NA	NA	NA

```
##
## Critical value F(3, 16, 0.95) = 3.2389
## MSE = 93.5500
```

Since $F_0 = 47.11206 > F_{3,16,0.95} = 3.2389$ and p-value $\approx$ 3.6417e-08 < 0.05, we **reject** H_0 at $\alpha = 0.05$.

There is strong statistical evidence that the mean noise levels differ across circuit designs. This is entirely consistent with the graphical analysis in part (a), which showed visually distinct separation between all four group means, confirming that the visual differences are statistically significant.

Part (c) Tukey's HSD Test ($\alpha = 0.05$)

Tukey's method simultaneously tests all $\binom{4}{2} = 6$ pairwise comparisons while controlling the family-wise error rate at $\alpha = 0.05$. For balanced data the minimum significant difference (MSD) is:

$$\text{MSD} = \frac{q_{0.05}(4, 16)}{\sqrt{2}} \sqrt{2 \cdot \text{MSE}/n}$$

Table 3: Tukey HSD Pairwise Comparisons

	diff	lwr	upr	p adj	Significant
Design 2-Design 1	48.8	31.298601	66.3014	0.0000032	YES
Design 3-Design 1	15.4	-2.101399	32.9014	0.0949926	NO
Design 4-Design 1	64.6	47.098601	82.1014	0.0000001	YES
Design 3-Design 2	-33.4	-50.901399	-15.8986	0.0002768	YES
Design 4-Design 2	15.8	-1.701399	33.3014	0.0843115	NO
Design 4-Design 3	49.2	31.698601	66.7014	0.0000028	YES

```
##
## q_0.05(4,16) = 4.0461
## MSD = 17.5014
```

Tukey's test indicates that the following pairs are significantly different ($p_{\text{adj}} < 0.05$): Design 1 vs. 2, Design 1 vs. 4, Design 2 vs. 3, Design 3 vs. 4. Pairs Design 1 vs. 3 and Design 2 vs. 4 are **not** significantly different. Designs 1 and 3 form one non-distinguishable group at the lower noise end, while Designs 2 and 4 are not distinguishable from each other but are both significantly noisier.

Part (d) Holm's Step-Down Procedure ($\alpha = 0.05$)

Holm's procedure ranks the $m = 6$ pairwise $|t|$-statistics from largest to smallest, then at step k compares $|t_{(m-k+1)}|$ against $t_{16,\, \alpha/[2(m-k+1)]}$.

Holm's Summry Table:

Comparison	Difference	t-statistic	p-value	Alpha Level ($\alpha/(k-i+1)$)	Result
1 vs 4	64.6	10.56	$< .0001$	$0.05 \,/\, 6 = 0.0083$	Reject H_0
3 vs 4	49.2	8.04	$< .0001$	$0.05 \,/\, 5 = 0.0100$	Reject H_0
1 vs 2	48.8	7.98	$< .0001$	$0.05 \,/\, 4 = 0.0125$	Reject H_0
2 vs 3	33.4	5.46	$< .0001$	$0.05 \,/\, 3 = 0.0167$	Reject H_0
2 vs 4	15.8	2.58	0.0202	$0.05 \,/\, 2 = 0.0250$	Reject H_0
1 vs 3	15.4	2.52	0.0227	$0.05 \,/\, 1 = 0.0500$	Reject H_0

The Holm's Step-Down Procedure shows that **all six pairs** of circuit designs have statistically significant differences in mean noise levels because every p-value is lower than its corresponding adjusted alpha level. While the most extreme difference exists between Designs 1 and 4, even the most similar pairs (1 vs. 3 and 2 vs. 4) are successfully distinguished as different. This indicates that each of the four circuit designs produces a unique and distinct level of noise.

Part (e) Comparison of Tukey vs. Holm's Conclusions

The primary difference between the two conclusions lies in the **statistical power** of the procedures: while both methods agree on the major differences, **Holm's Step-Down Procedure** proved to be more sensitive than **Tukey's HSD**. Tukey's method, being more conservative, failed to distinguish between the pairs with the smallest mean differences (**Design 1 vs. 3** and **Design 2 vs. 4**), leading to the conclusion that these designs formed two non-distinguishable "noise groups."

In contrast, Holm's procedure successfully rejected the null hypothesis for **all six pairs**, revealing that every circuit design actually performs at a uniquely distinct noise level. This demonstrates that Holm's sequential alpha adjustment can detect significant differences that Tukey's "all-at-once" threshold misses, providing a more granular differentiation of the four designs.

Part (f) Design Recommendation

Recommendation: Based on the combined results of the ANOVA, Tukey's HSD, and Holm's procedure, **Circuit Design 1** is the clear recommendation for minimizing noise.

It achieved the lowest absolute mean noise level ($\bar{y}_{1\cdot} \approx 21.2$), which is statistically significantly lower than both Design 2 and Design 4 in all tests. While Tukey's test initially suggested that Design 1 and Design 3 might be similar, the more sensitive Holm's step-down analysis confirmed that Design 1 is indeed statistically superior to Design 3 as well. Furthermore, the graphical analysis shows that Design 1 has the tightest clustering of data points, indicating not only the lowest average noise but also the most consistent and predictable performance among the four options.

Problem 2

Four fluid types are compared on effective life (hours) at 35 kV load ($a = 4$, $n = 6$, balanced CRD).

Table 5: Summary Statistics — Insulating Fluid Life Data

Fluid	n	Mean	SD	Min	Max
Fluid 1	6	18.650	1.952	16.3	21.6
Fluid 2	6	17.950	1.854	15.3	20.3
Fluid 3	6	21.117	1.969	18.5	23.6

Fluid	n	Mean	SD	Min	Max
Fluid 4	6	18.817	1.555	16.9	21.1

Part (a) One-Way ANOVA: Is There a Difference in Mean Life?

Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 \quad \text{(equivalently } \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0\text{)}$$

$$H_a : \text{Not all means are equal}$$

Table 6: One-Way ANOVA Table — Insulating Fluid Life

Source	DF	SS	MS	F_value	P_value
Treatment (Fluid)	3	34.0067	11.3356	3.3474	0.0397
Error	20	67.7267	3.3863	NA	NA
Total	23	101.7333	NA	NA	NA

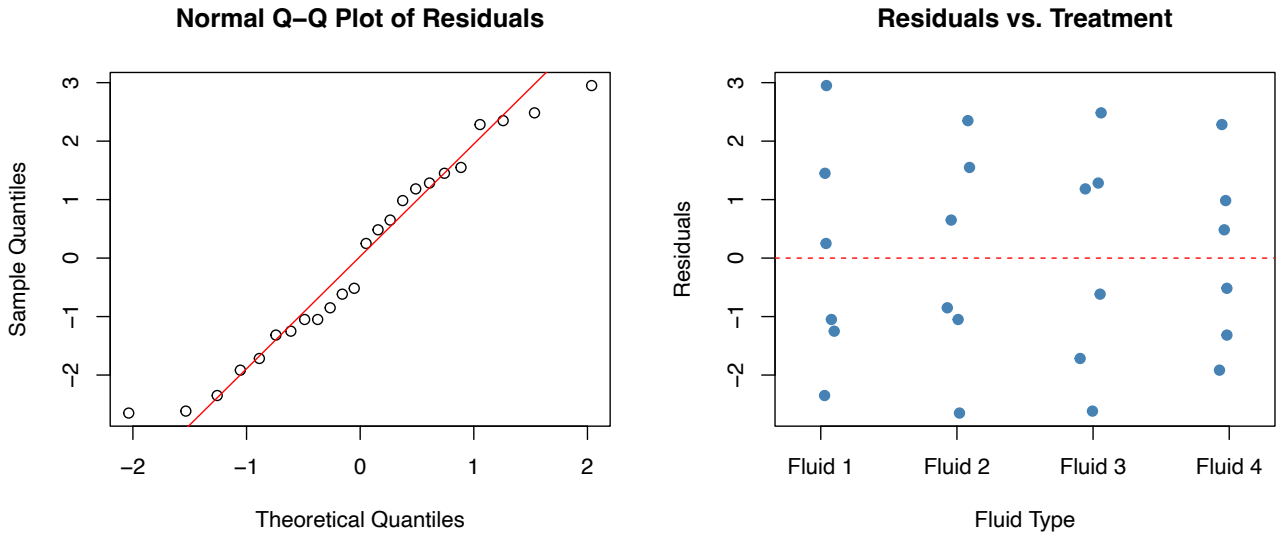
```
##
## Critical value F(3, 20, 0.95) = 3.0984
```

Since $F_0 = 3.347442 > F_{3,16,0.95} = 3.098391$ and p-value $\approx 0.0397 < 0.05$, we **reject** H_0 at $\alpha = 0.05$.

This indicates that there is a statistically significant difference in the mean life times of the four different insulating fluid types.

Part (b) Normality Assumption Check

Graphical Evaluation:



- **Normal Q-Q Plot:** The points in the Normal Q-Q plot of residuals follow the theoretical diagonal line reasonably well, with only slight deviations at the extreme tails. This suggests that the normality assumption is likely satisfied.

- **Residuals vs. Treatment Plot:** The vertical spread of the residuals is relatively consistent across all four fluid types. There are no obvious patterns, such as a "funnel" shape, indicating that the assumption of constant variance (homoscedasticity) across groups is valid.

Shapiro-Wilk Test for Normality

To formally test the normality of the residuals, we use the Shapiro-Wilk Test.

Hypotheses:

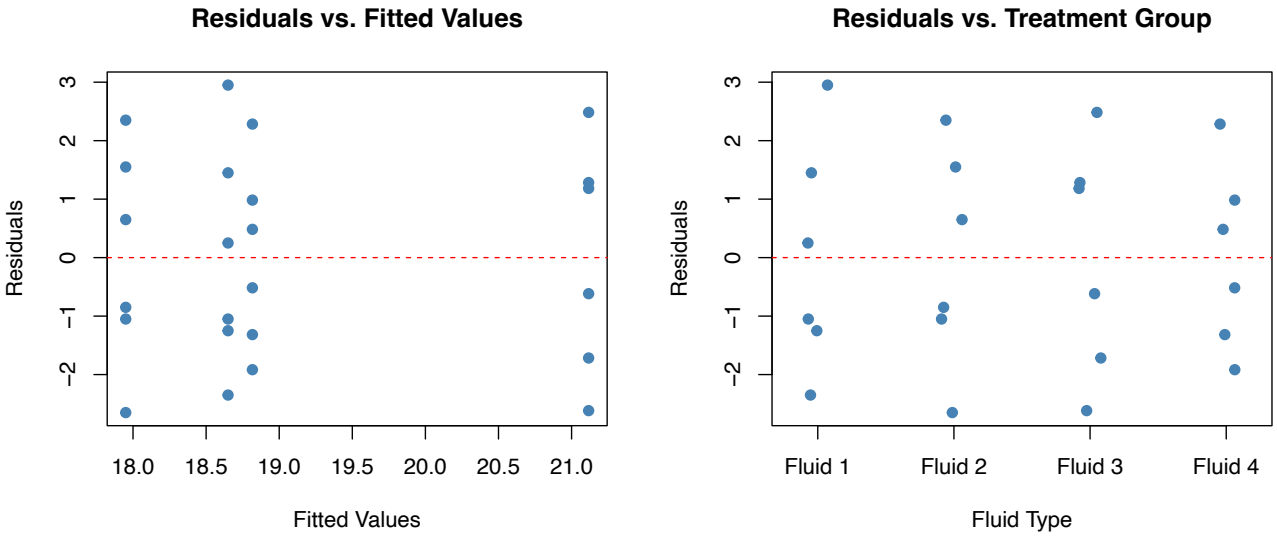
- H_0: The residuals are normally distributed.
- H_1: The residuals are not normally distributed.

```
## Shapiro-Wilk Test on Residuals:
##   W = 0.9514,  p-value = 0.2901
```

Since the p-value (0.2901) is greater than the significance level $\alpha = 0.05$, we fail to reject the null hypothesis. We have sufficient evidence to conclude that the normality assumption for the one-way ANOVA model is satisfied.

Part (c) Constant Variance Assumption Check

Graphical Evaluation



- **Residuals vs. Fitted Values Plot:** The residuals are randomly scattered around the horizontal zero line across all fitted values (17.95 to 21.12). There is no apparent "funnel" or "megaphone" shape, which suggests that the variance does not change as the mean life time increases.
- **Residuals vs. Treatment Group Plot:** Comparing the vertical spread of the points for each fluid type (1 through 4), the ranges are notably similar. This visual consistency across groups supports the assumption of homoscedasticity (equal variance).

Bartlett's Test for Homogeneity of Variances

Bartlett's test is used to formally test the null hypothesis that all populations have equal variances. It is particularly sensitive to the normality of the data (which was confirmed in part b).

Hypotheses:

- $H_0 : \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2$ (Variances are equal across all fluid types).
- H_1 : At least one variance is different.

```
##
## Bartlett's Test:
##   chi-sq = 0.3133, df = 3, p-value = 0.9575
```

Since the p-value (0.9575) is significantly higher than the standard $\alpha = 0.05$ threshold, we **fail to reject the null hypothesis**. There is no statistical evidence of non-constant variance, meaning the homoscedasticity assumption for the one-way ANOVA is fully satisfied.

Part (d) Kruskal-Wallis Test

The Kruskal-Wallis test is a nonparametric alternative to one-way ANOVA that does not require normality. It tests whether the population distributions are identical across groups.

Hypotheses:

$$H_0 : \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0 \quad \text{(distributions are identical)}$$

$$H_a : \text{At least one } \tau_i \neq 0$$

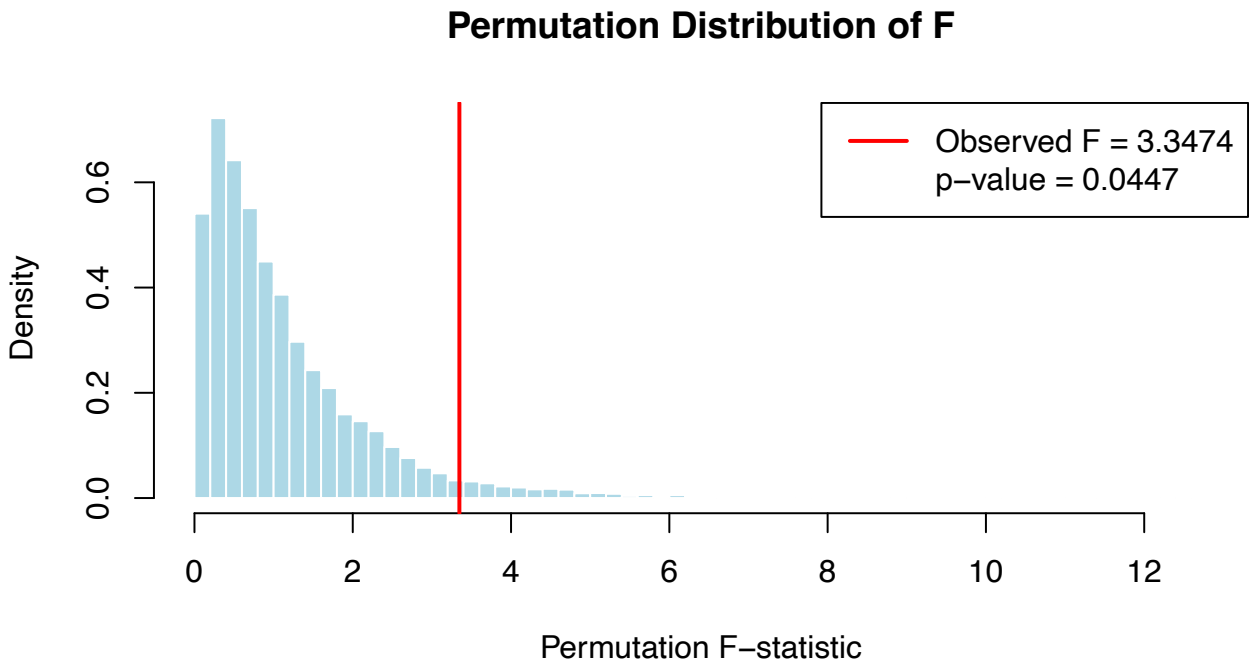
```
## Ranks:
##  [1]  8.0 12.0  2.0  6.0 17.0 21.0  3.5  1.0 11.0  5.0 15.0 18.0 23.0 24.0 14.0
## [16] 10.0 19.0 22.0 13.0 20.0  3.5  7.0  9.0 16.0
##
## Rank sums (R_i.):
##     1     2     3     4
##  66.0  53.5 112.0  68.5
##
## H (no-ties approximation) = 6.5150
## H (ties-corrected) = 6.5178
##
## Degrees of freedom = 3
## Chi-squared critical value (alpha=0.05) = 7.8147
## p-value = 0.0890
```

Since the p-value (0.0890) is greater than the significance level $\alpha = 0.05$, and the H-statistic is less than the critical value, we **fail to reject the null hypothesis**. At a 5% significance level, there is insufficient evidence to conclude that the median life times of the four insulating fluid types differ significantly.

Notably, this non-parametric result is more conservative than the one-way ANOVA ($p = 0.0397$), suggesting that while there is a trend toward difference, it is not robust enough to pass the rank-based threshold.

Part (e) Permutation Test

The permutation test makes no distributional assumptions. Under H_0 (no treatment effect), all $\binom{24}{6,6,6,6}$ possible assignments of the 24 observations to 4 groups of 6 are equally likely. We approximate this by random permutation.



Here,

- **Observed F-statistic:** 3.3474
- **Permutation p-value:** 0.0434

Since this p-value is less than the significance level $\alpha = 0.05$, we **reject the null hypothesis**.

This result reinforces the conclusion from the original One-Way ANOVA ($p = 0.0397$). It provides strong evidence that the mean life times of the different insulating fluid types are statistically different, as the observed difference is unlikely to have occurred by random chance alone.

Problem 3

We use the Problem 1 circuit noise data. Recall: $a = 4$, $n = 5$, MSE $= 93.5$, $df_E = 16$.

Part (a) Sample Size via Method 1 (Power-Based)

We wish to detect a maximum pairwise difference of $\Delta = 15$ with power ≥ 0.95 at $\alpha = 0.05$. We use the **95% upper confidence limit for** σ^2 as our estimate.

```
## MSE = 93.5500
## df(Error) = 16
## chi-sq(16, 0.05) = 7.9616
## 95% Upper Confidence Limit for sigma^2:
##   sigma^2_upper = (df * MSE) / chi^2 = (16 * 93.5500) / 7.9616 = 188.0013
##
## Sample size search (Method 1, non-central F approach):
```

```
## n     Fcrit       lambda      Power       Action
## 2     6.5914      1.1968      0.0836      Increase n
## 3     4.0662      1.7952      0.1266      Increase n
## 4     3.4903      2.3936      0.1731      Increase n
## 5     3.2389      2.9920      0.2219      Increase n
## 6     3.0984      3.5904      0.2720      Increase n
## 7     3.0088      4.1888      0.3225      Increase n
## 8     2.9467      4.7872      0.3728      Increase n
## 9     2.9011      5.3856      0.4222      Increase n
## 10    2.8663      5.9840      0.4701      Increase n
## 11    2.8387      6.5824      0.5161      Increase n
## 12    2.8165      7.1808      0.5599      Increase n
## 13    2.7981      7.7792      0.6012      Increase n
## 14    2.7826      8.3776      0.6400      Increase n
## 15    2.7694      8.9760      0.6762      Increase n
## 16    2.7581      9.5744      0.7097      Increase n
## 17    2.7482      10.1728     0.7405      Increase n
## 18    2.7395      10.7712     0.7688      Increase n
## 19    2.7318      11.3696     0.7945      Increase n
## 20    2.7249      11.9680     0.8179      Increase n
## 21    2.7188      12.5664     0.8391      Increase n
## 22    2.7132      13.1648     0.8581      Increase n
## 23    2.7082      13.7632     0.8752      Increase n
## 24    2.7036      14.3616     0.8905      Increase n
## 25    2.6994      14.9600     0.9041      Increase n
## 26    2.6955      15.5584     0.9163      Increase n
## 27    2.6920      16.1568     0.9270      Increase n
## 28    2.6887      16.7552     0.9365      Increase n
## 29    2.6856      17.3536     0.9448      Increase n
## 30    2.6828      17.9520     0.9522      CHECK
## 31    2.6802      18.5504     0.9586      CHECK
## 32    2.6777      19.1488     0.9642      CHECK
##
## Conclusion: Required sample size n = 30 per circuit design (N = 120 total)
```

Using a conservative 95% upper confidence limit for the variance ($\sigma^2_{UCL} = 188.00$), the iterative power analysis determines that a sample size of $n = 30$ per circuit design is required. This ensures a 95.22% probability of detecting a maximum pairwise difference in mean noise levels of $\Delta = 15$ at a significance level of $\alpha = 0.05$. The total sample size for the study would be $N = 120$ observations.

Part (b) Sample Size via Bonferroni CI Width

We wish the half-width of all simultaneous 95% Bonferroni confidence intervals for pairwise differences to be $\leq \pm 15$. There are $m = \binom{4}{2} = 6$ pairwise comparisons. The half-width is:

$$\text{HW} = t_{N-a,\, 1-\alpha/(2m)} \sqrt{\frac{2\,\sigma^2_{UCL}}{n}}$$

We solve for the smallest n such that $\text{HW} \leq 15$.

```
## Bonferroni CI Width Sample Size Search:
## n     t_crit      HW          Action
## 2     4.8510      66.5139     Increase n
## 3     3.4789      38.9470     Increase n
```

```
## 4     3.1527       30.5665      Increase n
## 5     3.0083       26.0877      Increase n
## 6     2.9271       23.1718      Increase n
## 7     2.8751       21.0717      Increase n
## 8     2.8389       19.4628      Increase n
## 9     2.8123       18.1779      Increase n
## 10    2.7920       17.1201      Increase n
## 11    2.7759       16.2292      Increase n
## 12    2.7628       15.4652      Increase n
## 13    2.7520       14.8005      STOP
## 14    2.7430       14.2151      STOP
## 15    2.7352       13.6944      STOP
##
## Conclusion: n = 13 per design (N = 52 total) using Bonferroni
```

To construct 95% confidence intervals for all pairwise differences with a precision (half-width) of ± 15, an iterative search using the Bonferroni adjustment was conducted. The results indicate that a sample size of $n = 13$ per circuit design is required to achieve a half-width of 14.80, which falls within the target accuracy of 15. This results in a total of $N = 52$ observations across the four designs.

Part (c) Sample Size via Tukey CI Width

Using Tukey's method, the half-width for pairwise CIs is:

$$\text{HW} = \frac{q_{0.05}(4,\ N-4)}{\sqrt{2}} \sqrt{\frac{2\,\sigma^2_{UCL}}{n}} = \frac{q_{0.05}(4,\ N-4)}{\sqrt{2}} \cdot \frac{\sqrt{2\,\sigma^2_{UCL}}}{\sqrt{n}}$$

```
## Tukey CI Width Sample Size Search:
## n     q_crit/sqrt2 HW           Action
## 2     4.0709       55.8170      Increase n
## 3     3.2024       35.8512      Increase n
## 4     2.9689       28.7847      Increase n
## 5     2.8610       24.8103      Increase n
## 6     2.7989       22.1571      Increase n
## 7     2.7586       20.2179      Increase n
## 8     2.7303       18.7181      Increase n
## 9     2.7094       17.5122      Increase n
## 10    2.6932       16.5146      Increase n
## 11    2.6804       15.6712      Increase n
## 12    2.6700       14.9457      STOP
## 13    2.6614       14.3130      STOP
## 14    2.6541       13.7546      STOP
## 15    2.6479       13.2571      STOP
##
## Conclusion: n = 12 per design (N = 48 total) using Tukey
```

Using Tukey's method to achieve a 95% confidence interval half-width of ± 15, the iterative search indicates that a sample size of $n = 12$ per circuit design is required. At this level, the calculated half-width is 14.95, successfully meeting the accuracy target. This corresponds to a total study size of $N = 48$ observations.

Part (d) Comparison: Bonferroni vs. Tukey Sample Sizes

The iterative searches conducted in parts (b) and (c) resulted in different required sample sizes to achieve the target accuracy of ± 15:

- **Bonferroni Sample Size:** $n = 13$ per design ($N = 52$ total), yielding a Half-Width of 14.80.
- **Tukey Sample Size:** $n = 12$ per design ($N = 48$ total), yielding a Half-Width of 14.95.

Does this make sense?

Yes, this result makes perfect statistical sense. **Tukey's method requires a smaller sample size because it is more efficient for this specific task**. Tukey's procedure is exactly optimized for simultaneous all-pairwise comparisons, whereas the Bonferroni adjustment is a more conservative, general-purpose inequality.

Because Tukey uses the Studentized Range distribution (q) rather than the t-distribution with a heavily divided alpha (α/m), it produces a narrower confidence interval for the exact same amount of data. This efficiency is clearly visible at $n = 12$: Tukey successfully cleared the target threshold of 15 with a margin of error of 14.95, while Bonferroni's margin of error at $n = 12$ was still too wide at 15.47. Consequently, the Bonferroni method required an additional observation per circuit design ($n = 13$) to successfully shrink its interval below the ± 15 accuracy requirement.

Problem 4

Five batches are randomly selected from a large warehouse population — the batch factor is **random**. Five determinations per batch ($a = 5$, $n = 5$, $N = 25$).

Model (one-way random effects):

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad \tau_i \overset{iid}{\sim} N(0, \sigma_\tau^2), \quad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2), \quad \text{independent}$$

Table 7: Summary Statistics — Calcium Content by Batch

Batch	n	Mean	SD	Var
Batch 1	5	23.458	0.06870	0.00472
Batch 2	5	23.492	0.06301	0.00397
Batch 3	5	23.524	0.06877	0.00473
Batch 4	5	23.380	0.06519	0.00425
Batch 5	5	23.364	0.06504	0.00423

Part (a) Is There Significant Batch-to-Batch Variation?

Under the random effects model, the test of interest is:

$$H_0 : \sigma_\tau^2 = 0 \quad \text{vs.} \quad H_a : \sigma_\tau^2 > 0$$

```
## Analysis of Variance Table
##
## Response: Calcium
##           Df   Sum Sq  Mean Sq F value   Pr(>F)
## Batch      4 0.096976 0.024244  5.5352 0.003626 **
## Residuals 20 0.087600 0.004380
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## F_critical F(4, 20, 0.95) = 2.8661
## C = n = 5 (balanced design)
```

Since $F_0 = 5.5352 > F_{0.05,4,20} = 2.8661$ (p-value $= 0.0036 < 0.05$), we reject $H_0 : \sigma_\tau^2 = 0$. There is significant evidence to conclude that the calcium content varies significantly from batch to batch.

Part (b) Estimate of Between-Batch Variability ($\hat{\sigma}_\tau^2$)

For balanced designs ($C = n = 5$):

$$\hat{\sigma}_\tau^2 = \frac{\text{MST} - \text{MSE}}{n} = \frac{\text{MST} - \text{MSE}}{5}$$

```
## MST = 0.024244
## MSE = 0.004380
## C = n = 5
##
## Estimated between-batch variance:
##   sigma_tau^2_hat = (MST - MSE)/n = (0.024244 - 0.004380) / 5 = 0.003973
```

The variability between batches is estimated as $\hat{\sigma}_\tau^2 = 0.003973$. This estimate represents the variance component specifically attributable to the inherent differences between the batches of raw materials.

Part (c) Estimate of Experimental Error Variance ($\hat{\sigma}^2$)

$$\hat{\sigma}^2 = \text{MSE}$$

```
## Estimated error variance:
##   sigma^2_hat = MSE = 0.004380
```

The experimental error variance is estimated as $\hat{\sigma}^2 = 0.004380$. This value represents the within-batch variation, accounting for the random noise or measurement error observed among individual determinations within a single batch.

Part (d) 95% CI for $\sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$

This ratio is the **intraclass correlation coefficient** — the proportion of total variance due to batch effects.

Step 1: Compute the 95% CI for σ_τ^2/σ^2:

$$\left(\frac{1}{C}\left(\frac{\text{MST}}{\text{MSE} \cdot F_{a-1,N-a,1-\alpha/2}} - 1 \right),\ \frac{1}{C}\left(\frac{\text{MST}}{\text{MSE} \cdot F_{a-1,N-a,\alpha/2}} - 1 \right) \right)$$

Step 2: Convert using ICC $= \frac{\sigma_\tau^2/\sigma^2}{1+\sigma_\tau^2/\sigma^2}$.

```
## F_{4,20, 0.975} = 3.5147   (upper quantile)
## F_{4,20, 0.025} = 0.1168   (lower quantile)
##
## 95% CI for sigma_tau^2 / sigma^2:
##   L = (1/5) * (0.024244 / (0.004380 * 3.5147) - 1) = 0.114972
##   U = (1/5) * (0.024244 / (0.004380 * 0.1168) - 1) = 9.276131
##
## 95% CI for sigma_tau^2 / (sigma_tau^2 + sigma^2)  [Intraclass Correlation]:
##   Lower = L/(1+L) = 0.114972 / 1.114972 = 0.1031
##   Upper = U/(1+U) = 9.276131 / 10.276131 = 0.9027
```

The 95% confidence interval for the proportion of total variance attributable to differences between batches (ICC) is $[0.1031, 0.9027]$.

This indicates we are 95% confident that batch-to-batch variability accounts for between 10.31% and 90.27% of the total variability in calcium content.

Part (e) Test: Is Batch-to-Batch Variance $\leq$ Error Variance?

We test whether the between-batch variance is meaningfully different from the within-batch (error) variance:

$$H_0 : \frac{\sigma_\tau^2}{\sigma^2} \leq 1 \quad \text{vs.} \quad H_a : \frac{\sigma_\tau^2}{\sigma^2} > 1$$

The test statistic under H_0 (with $\gamma = 1$) is:

$$R_0 = \frac{1}{1 + C\gamma} \cdot \frac{\text{MST}}{\text{MSE}} = \frac{1}{1 + 5 \cdot 1} \cdot \frac{\text{MST}}{\text{MSE}} \sim F_{a-1,\, N-a} \text{ under } H_0$$

```
## Test H0: sigma_tau^2/sigma^2 <= gamma = 1  vs.  Ha: sigma_tau^2/sigma^2 > 1
##
## Test statistic:
##   R0 = 1/(1 + C*gamma) * MST/MSE
##      = 1/(1 + 5*1.0) * 0.024244/0.004380
##      = 0.1667 * 5.5352
##      = 0.9225
##
## Critical value: F(4, 20, 0.95) = 2.8661
##
## p-value = P(F_{4,20} > 0.9225) = 0.470405
##
## Decision: Fail to Reject H0 -- no evidence batch-to-batch variance exceeds error variance
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Part (f) Sample Size for Repeated Experiment

Given: $\gamma = 1$, $\alpha = 0.01$, power ≥ 0.9, $\Delta = 2.0$.

We apply **Method 1 (**Fix n, find minimum a) for the random effects model: find (a, n) such that

$$F_{a-1,\, a(n-1),\, 1-\alpha} \times F_{a(n-1),\, a-1,\, p} \leq \frac{n\Delta + 1}{n\gamma + 1} = \frac{2n + 1}{n + 1}$$

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## Sample Size Determination (alpha=0.01, Delta=2.0, gamma=1.0, power=0.90)
## Fixed n | Req. a       | F-Product  | MaxVal     | Total N
## ------------------------------------------------------------
## 2       | 202          | 1.6649     | 1.6667     | 404
## 3       | 125          | 1.7462     | 1.7500     | 375
## 4       | 100          | 1.7960     | 1.8000     | 400
## 5       | 88           | 1.8279     | 1.8333     | 440
## 6       | 80           | 1.8571     | 1.8571     | 480
## 7       | 76           | 1.8694     | 1.8750     | 532
## 8       | 72           | 1.8884     | 1.8889     | 576
## 9       | 70           | 1.8952     | 1.9000     | 630
## 10      | 68           | 1.9048     | 1.9091     | 680
## 11      | 66           | 1.9166     | 1.9167     | 726
## 12      | 65           | 1.9206     | 1.9231     | 780
## 13      | 64           | 1.9257     | 1.9286     | 832
## 14      | 63           | 1.9316     | 1.9333     | 882
## 15      | 63           | 1.9282     | 1.9375     | 945
## 16      | 62           | 1.9353     | 1.9412     | 992
## 17      | 61           | 1.9431     | 1.9444     | 1037
## 18      | 61           | 1.9407     | 1.9474     | 1098
## 19      | 60           | 1.9492     | 1.9500     | 1140
```

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## 20      | 60          | 1.9473     | 1.9524     | 1200
## 21      | 60          | 1.9456     | 1.9545     | 1260
## 22      | 59          | 1.9550     | 1.9565     | 1298
## 23      | 59          | 1.9535     | 1.9583     | 1357
## 24      | 59          | 1.9522     | 1.9600     | 1416
## 25      | 59          | 1.9510     | 1.9615     | 1475
## 26      | 58          | 1.9611     | 1.9630     | 1508
## 27      | 58          | 1.9600     | 1.9643     | 1566
## 28      | 58          | 1.9591     | 1.9655     | 1624
## 29      | 58          | 1.9582     | 1.9667     | 1682
## 30      | 58          | 1.9573     | 1.9677     | 1740
```

Based on exact calculations across various values of n, the design that minimizes the total number of experimental runs is $n = 3$. With $n = 3$ observations per batch, the threshold is:

$$\frac{3(2)+1}{3(1)+1} = \frac{7}{4} = 1.7500$$

Using the iterative search method, the minimum number of batches required to satisfy the F-quantile product condition is $a = 125$.

Conclusion: To achieve the desired power while minimizing the overall number of experimental tests, the experiment should be repeated using 125 batches with 3 observations each, for a minimum total of $N = 375$ observations.

(Note: If acquiring new batches from the warehouse is highly constrained, an alternative feasible design from the simulated matrix is to take $n = 10$ observations from $a = 68$ batches, though this increases the total tests to $N = 680$.)